

[Lecture Note] Analyzing Quantitative Data

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Learning Objectives

- 10-1 Summarize why statistics are used in research methods
- 10-2 Conduct univariate analysis
- 10-3 Explain the measures of central tendency
- 10-4 Define measures of variability and dispersion
- 10-5 Describe the various ways to graphically represent data
- 10-6 Conduct bivariate analysis

SHORT VIDEO INTRODUCTION

<https://www.youtube.com/watch?v=KU0O7fnkjXU>

<https://www.youtube.com/watch?v=EUeQRE5UJpg>

Analyzing Quantitative Data

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- Statistics: a branch of mathematics that uses mathematical logic to interpret phenomena and draw conclusions from the accumulated data.
 - It establishes relationships between and among variables and can even attempt to predict future behaviors or relationships.

Univariate Analysis

- Univariate analysis: the analysis of only one variable.
- Descriptive analysis: allows researchers to sketch the details of variables and gain familiarity with the sample; gives an idea of the descriptive characteristics of variables.
 - Descriptive statistics describe the first impressions of a variable.
- Inferential analysis/statistics: provide a deeper understanding of the variables because it draws conclusions about the population from the sample at hand.
 - This type of analysis goes beyond the description of the data available and attempts to draw relationships or associations between variables.
 - Once a relationship is detected between variables, inferential statistics attempts to predict the possibility of applying the same relationship to the entire population of the study.

- Inferential statistics operate by trying to calculate the random error that occurs during collection and analysis.
- Random error: associated with the type of common mistakes or errors that happen by chance when data are collected or entered into a spreadsheet or during analysis.
- Sampling error: a random error that occurs during the process of data collection.
 - Inferential statistics only investigates the sampling error that can happen randomly; it is unable to identify a systematic sampling error.

Univariate analysis is the first step researchers take to familiarize themselves with the population sample.

- Frequency Distributions
 - Constructing a frequency distribution with data is a simple procedure that can illustrate key characteristics of participants in the sample.
 - The first indicator is absolute frequency, which reflects the number of times an event or instance has been repeated.
 - Assigning percentages to each category adds another layer of valuable information.
 - * Percentages are capable of zooming out from the raw numbers because it compares the raw numbers to the entire sampling population.
 - Cumulative percent: another descriptive measure of frequency; shows the collective percentage of categories together.
 - * It is sensitive to the order of organization.

Measure of Central Tendency

Measures of central tendency: summarize an entire variable or category in a single value that is numerically representative of it.

- Mean: the mathematical average of a set of numbers; all numbers in a variable category are added together and then divided by n , which is the total number of entries in that data set.
 - The mean is useful for comparing two or more groups.
 - The mean takes into account every single point in the variable, and it is very sensitive to outliers.
 - Outlier: a data point that is much higher or lower than the majority of other points.
- Median: represents the center point in a set of numbers that have been arranged in ascending or descending order.
 - The median is useful in statistics because it gives an understanding about the distribution of a variable.
- Mode: the most frequently used number in a data set; which number is most commonly encountered in a specific set of numbers.
 - The mode can provide additional information that can potentially clarify any misleading information from the mean or median.
 - One variable can have more than one mode.
 - * Unimodal: only one mode
 - * Bimodal: two modes
 - * Multimodal: three or more modes

Measures of Variability and Dispersion

- Measures of central tendency tell about the middle scores in a distribution, but they tell very little about how they are spread out.
- Measures of variability and dispersion are the ones that give information about how far apart or close by the values are in a distribution.
- Variability expresses the degree to which data in a data set spreads out, or deviates, from the center.

- Homogenous: sets that have little variability
- Heterogenous: sets with a lot of variability
 - Pose a higher level of complexity in approaching analysis.

- Range: the difference between the largest and smallest numbers in the set.
 - Interquartile range (IQR) is another measure of the middle 50% of the scores in a variable
- Variance: represents each datapoint's average deviation from the mean; the average sum of the squared distance of every single point in the data set from the mean.
- Standard deviation: the square root of the variance; also represents each data point's average deviation from the mean.

Normal Distribution

- The normal distribution is a theoretical perfect distribution where the mean, mode, and median are equal to one another.
 - It is symmetrical; the mean divides the distribution into equal halves.
 - It is used to compare against the actual distribution.
- Standard scores: also called z scores; indicate the number of standard deviations in a distribution and what proportion of the scores fall within this range.
- The proportion of z scores that fall between z scores in any normal distribution, regardless of its means or standard deviations, are defined by the following rules:
 - The area between z scores 0 and -1 and between 0 and 1 always equal 0.3413.
 - The area between z scores 1 and 2 and between -1 and -2 always equal 0.136.
 - The area between z scores 2 and 3 and between -2 and -3 always equal .0215.
 - The proportion of scores greater than z score 3 or smaller than z score -3 is 0.0013.
- These rules mean that approximately 68% of the data points fall between one standard deviation below and above the means.
 - 99% fall between 2 standard deviations below and above the mean.

Skewness and Kurtosis

- Skewness: when a distribution lacks the symmetry created when the mean, median, and mode are in the center; its shape is asymmetric when viewed in a histogram.
 - The term used to note that a distribution is clustered on the right or the left.
 - A distribution with a long tail on the left side and a majority of data on the right is called left-skew, left-tailed, or negatively skewed distribution.
 - A distribution with a long tail on the right but a majority of points of the left is called a right-skew, right-tailed, or positively skewed distribution.
 - In right/positively skewed distributions, the mode will be on the left of the distribution where the highest peak is, the median will have a greater value than the mode, and the mean will have a higher value than the median and mode.
 - The opposite occurs in left/negatively skewed distributions; the mean will have the smallest value, followed by the median, and then the mode, which will be at the highest peak.

- Kurtosis: refers to the flatness of the distribution and the flatness of its tails.
 - The two characteristics work together; a high peak in the distribution also has long flat tails.
 - * Leptokurtic: distribution with a high peak and long flat tails
 - * Platykurtic: distribution with a flat peak and short tails.
 - Kurtosis is an indication that some problematic outliers are present in the distribution.
 - If kurtosis is noted during analysis, the researcher will often take another look at the distribution to possibly remove outliers and bring the distribution to a shape closer to normal distribution.

Continuous and Discrete Variables

- Discrete variables: categorical variables that can only take whole numbers and do not have any continuity in terms of measurement.
 - Example: number of employees in an institution
- Continuous variable: a type of numerical variable that can take all possible units
 - Examples: income, height, weight

V. Graphical Representation of Data

A graphic representation of information can have a powerful impact on a broader audience, visually depicting relationships and patterns among variables.

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- Bar graphs: use vertical bars that represent each variable in a study.
 - The viewer can easily distinguish not only which category occurs most frequently, but how it compares to the other categories.
 - Histograms: often used for quantitative variables where the categories are related to one another.
 - Presents the frequency of a variable in the form of distinct but connected bars.
 - Line graph: a line connecting points in a distribution.
 - Useful in depicting two or more categories simultaneously, or if there are multiple distributions in the same category.
 - A line graph is an effective way to demonstrate relationships between variables.
 - Pie charts: a visual representation of the distribution of one variable.
 - Work best with categorical variables that have a small number of categories.
 - The main advantage of a pie chart is its visual impact on the audience.

VI. Bivariate Analysis

- Bivariate analysis: used to investigate the relationship between two variables; particularly, to understand whether the changes in one variable are reflected by changes in another.
- Correlation: used to determine the extent to which two quantitative variables approximate a linear relationship.
- Regression: used to identify the line that best describes this relationship.
- The Linear Model
 - i. A linear relationship between variables is a perfect relationship between two variables that, when graphed, resembles a straight line.
 - ii. In a linear relationship, the changes in units of one variable are reflected in specific interval changes in the other variable.
 - iii. They are rare in research findings because most of the variables collected from participants will have variability in the answers.

Correlation

- When all the data are cleaned and entered in a software program, scientists will typically run an analysis using a Pearson correlation coefficient, called r —an index that determines the level of correlation between two variables
- Pearson correlation coefficient r can range in value from -1 to 1 , indicating the level of linearity between two variables.
 - The closer this coefficient, r , is to -1 or 1 , the more likely it is that the relationship resembles a linear relationship.
 - The closer this coefficient is to 0 , the less likely it is that these two variables are correlated to each other.
 - Correlations of 1 or -1 are perfect positive and negative relationships, respectively, while a correlation coefficient of 0 indicates no relationship between variables.

Causation

- The fact that two variables are strongly correlated with one another does not mean that one variable causes changes in the other variable.
- A spurious relationship happens when the correlation between two variables is affected by a third variable that interferes with the relationship we are studying.

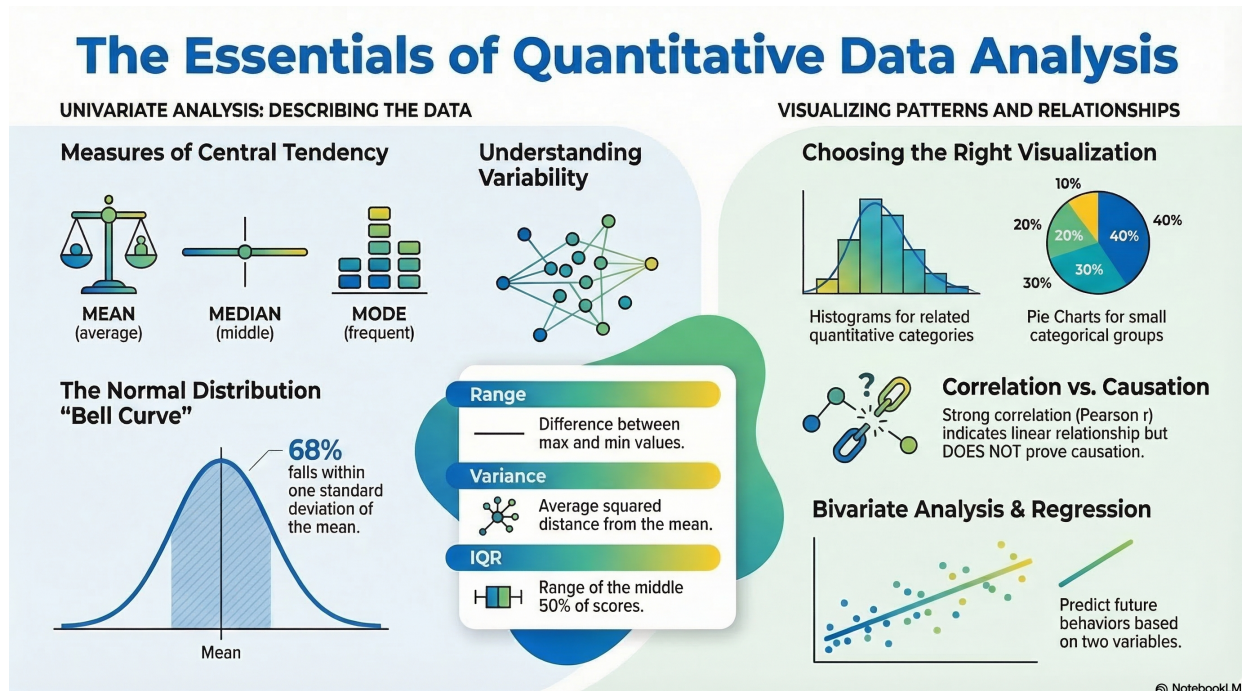
Regression

- Regression/best fitting line: represents the closest possible linear relationship between two variables.
- Regression is used to predict future behaviors of the variables as much as to explain current behaviors.
 - The best fitting line of the relationship between variables also tells us the direction of a relationship

Scatterplots

- Scatterplots visually depict a bivariate relationship.
- Scatterplots are dots on a graph that represent a relationship between two quantitative variables: dependent and independent.
 - Each single dot represents one case.
 - The dependent variable is on the vertical axis (or y-axis) and the independent variable on the horizontal axis (or x-axis).

Chapter Infographic



NotebookLM 생성 요약 인포그래픽. 해설 영상·팟캐스트는 [resource/](#) 참조.

Quiz

출처: 02_Assets/assessment/GE2021_TestBank.xlsx (chapter=10) · 5문항

Q1. Univariate analysis examines only one variable at a time.

- A. TRUE
- B. FALSE

i 정답 보기

정답: A

Univariate analysis focuses on a single variable, while bivariate analysis examines the relationship between two variables.

Q2. Statistics is BEST defined as:

- A. A branch of biology that studies populations
- B. A type of qualitative coding system
- C. A branch of mathematics that uses logic to interpret phenomena and draw conclusions from accumulated data
- D. A scale used in surveys

i 정답 보기

정답: C

Statistics is the mathematical discipline used to interpret data and draw conclusions about variables and relationships.

Q3. Which statement BEST distinguishes descriptive from inferential statistics?

- A. Descriptive analyzes the population; inferential only the sample
- B. Descriptive uses no math; inferential uses no math
- C. Descriptive is qualitative only; inferential is quantitative only
- D. Descriptive describes the sample at hand; inferential draws conclusions about the population from the sample

i 정답 보기

정답: D

Descriptive statistics summarize the sample, while inferential statistics generalize to the broader population.

218 **Q4. “Absolute frequency” in a frequency distribution refers to:**

- 219 • **A.** The percentage of cases that fall in a category
- 220 • **B.** A cumulative percentage
- 221 • **C.** The raw count of how many times an event or category occurs
- 222 • **D.** The standard deviation of the distribution

i 정답 보기

정답: **C**

Absolute frequency is the raw number of times something occurs (e.g., 34 high-school graduates).

224 **Q5. For the dataset 2, 4, 6, 8, 20, 24, 30, 32, 34, 40, the median is:**

- 225 • **A.** 20
- 226 • **B.** 22
- 227 • **C.** 24
- 228 • **D.** 30

i 정답 보기

정답: **B**

With an even number of values, the median is the average of the two middle numbers $(20 + 24)/2 = 22$.

230 **Reference**